

Dipeolu Ayomide Michael

Python Engineer | AI & ML Systems | Backend APIs | LLM Integration | Data Pipelines

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🔗 dipeolu-ayomide-main-port.vercel.app

Profile

Python Engineer with a strong background in AI systems, ML engineering, and scalable backend development. Experienced in building and deploying LLM-powered applications across GCP, AWS, and Azure, with practical expertise in LLMOps, Docker-based containerization, and AI product scaling using concurrency control. Has delivered production-grade RESTful APIs, SQL data pipelines, and AI agent integrations for internal tools and automation platforms. Brings a full-stack engineering mindset from model integration and cloud infrastructure to CI/CD pipelines and cross-platform API development.

Professional Experience

2026/01 – present
Lagos, Nigeria

AI Engineer

MacTay Consulting (Rope Africa) 

- Architected the AISA Omni-Channel Ecosystem by engineering a suite of conversational AI and voice agents leveraging RAG and intelligent pause-handling to automate customer engagement across web platforms and LMS environments.
- Engineered an Enterprise Agentic Email System for Gmail and Outlook featuring hybrid manual/auto-pilot modes and an OCR-enabled knowledge base supporting PDF, DOCX, and CSV ingestion successfully demonstrated to First Bank and Altbank.
- Developed a Multilingual AI QA Pipeline to analyse and score 5,000+ calls for sentiment and responsiveness, integrating cloud storage across OneDrive, Google Drive, and SharePoint with support for French, Portuguese, Chinese, and Spanish as well as local dialects like Yoruba, Igbo and Hausa.
- Built a Dual-Interface Call Centre Support System with an agent-facing chat interface for real-time assistance and a dedicated admin panel for knowledge base management and monitoring.

2025/12 – 2026/03
Remote

Python Backend Developer

RaterX Africa · 

- Architected and built a production REST API (FastAPI/Python) powering a live talent marketplace platform, covering authentication, talent profiling, pilot requests, admin tooling, assessments, and a full blog CMS across 7 modules.
- Implemented JWT and OTP authentication with role-based access control (RBAC) for Admin and Talent roles, including a secure anti-enumeration password reset flow.
- Integrated Tally webhooks with HMAC-SHA256 signature verification and auto-tiering logic to automatically classify talents based on assessment scores, with configurable retake cooldown enforcement.
- Built file management pipelines using Google Cloud Storage for private resume uploads with 15-minute expiring signed URLs and public profile picture hosting.
- Designed an anonymous engagement system using IP and User-Agent fingerprinting to deduplicate blog likes and views without requiring authentication.
- Implemented an async email subscription system with HMAC-signed unsubscribe tokens and background notifications on new blog post publish, using Jinja2 HTML email templates.

- Dockerized the full application with docker-compose, configured Alembic migrations, and used async PostgreSQL via asyncpg and SQLAlchemy 2.0 throughout.
- Platform is live and actively serves thousands of registered talents globally.

2025/09 – 2025/11
Remote

AI/ML Engineer — Steganography & Product Authentication

Rivabit 

- Researched steganography techniques and applied findings to build a product authentication system for pharmaceuticals, beverages, and FMCG goods.
- Built a QR code encoder using the Steno Python library to embed unique, invisible identifiers into product packaging for originality verification.
- Developed a custom decoder using the ZXing Java library capable of authenticating encoded QR codes and distinguishing genuine products from counterfeits.
- Reviewed and synthesised multiple academic papers on steganographic encoding approaches to determine the most robust and scalable technique for production use.

2024/10 – 2026/04
Lagos, Nigeria

AI/ML Engineer

Optimus AI Labs 

- Designed and deployed an AI-powered dental phone agent using Twilio, enabling 24/7 autonomous patient service and reducing front desk workload by approximately 80%.
- Architected a hybrid RAG knowledge base using PostgreSQL and Pinecone with advanced re-ranking techniques, improving response accuracy by 70% and reducing query latency by 60%.
- Engineered multi-agent AI workflows with LangGraph and LangChain, incorporating validation layers and feedback loops to ensure reliable production performance.
- Managed the full LLM lifecycle from fine-tuning with PyTorch/TensorFlow to production deployment implementing token optimisation strategies that reduced inference costs by 25%.
- Researched and validated multiple RAG architectures (hybrid, multi-query, recursive retrieval) to determine optimal strategies per use case, improving system relevance by up to 40%.
- Developed scalable backend APIs with FastAPI/Flask and PostgreSQL/MySQL to support AI services, and automated data pipelines for web scraping and preprocessing using Pandas.
- Deployed and optimised AI solutions on Railway and Runpod within Linux environments, ensuring scalable and reliable cloud operations.

2023/05 – 2023/11
Remote

Data Science Intern

Hamoye AI Labs 

- Developed custom web scrapers to collect structured and unstructured data from diverse online sources for downstream ML pipelines.
- Cleaned and transformed raw datasets to support accurate model training and exploratory data analysis.
- Performed EDA to surface trends and inform data-driven decision-making across multiple project domains.
- Built and evaluated classification, regression, and clustering models, optimising for performance across varied tasks including time series analysis.

Technical Skills

AI & ML Engineering

LLM Integration, LLMOps, RAG (Hybrid, Multi-query, Recursive), LangChain, LangGraph, Prompt Engineering, Multi-agent Workflows, Fine-tuning (PyTorch / TensorFlow), Vector Databases (Pinecone, ChromaDB, PGVector, FAISS), Conversational AI, Voice Agents (Twilio), Claude, GPT, Gemini, Grok, Cohere

Databases & Data Engineering

PostgreSQL, MySQL, SQLite, MongoDB, asyncpg, Pandas, NumPy, Query Optimisation, Schema Design, Data Indexing, Web Scraping, Data Pipelines

Integrations & Tooling

Google Drive API, Microsoft OneDrive, SharePoint, Google Cloud Storage, Google Speech-to-Text / TTS, Tally Webhooks, Resend, Google Sheets API, OCR

Data Science & Analysis

Exploratory Data Analysis (EDA), Classification & Regression Models, Clustering, Time Series Analysis, Sentiment Analysis, ML Model Deployment

Backend Development

Python, FastAPI, Flask, RESTful APIs, SQLAlchemy (async), Alembic, JWT & RBAC, Webhook Integration, Jinja2, Microservices Architecture, Modular System Design

Cloud & Infrastructure

Google Cloud Platform (GCP), AWS, Microsoft Azure, Docker, docker-compose, Kubernetes (basic), Railway, Runpod, GitHub Actions, CI/CD Pipelines, Linux / Bash

Security & Specialised Techniques

Steganography (Steno / ZXing), QR Code Authentication, HMAC-SHA256, Anti-enumeration Design, IP Fingerprinting, Signed URL Management

Featured Projects

FPL Padi — AI Fantasy Premier League Assistant

FastAPI · LangGraph · GPT-4o-mini · Gemini · Redis · PostgreSQL · Docker · Railway

- Built a conversational AI backend for Fantasy Premier League management using FastAPI and LangGraph, enabling natural language squad recommendations and player analysis powered by GPT-4o-mini.
- Implemented a natural language to SQL pipeline using Gemini 2.5 Flash to query a live database of 500+ players across 80+ attributes including xG, form, fixture difficulty, and cost metrics.
- Engineered an image analysis feature using Gemini 2.0 Flash Vision to extract structured team data, players, costs, formation, and injuries directly from FPL screenshots.
- Automated daily web scraping pipelines with Playwright and APScheduler to keep player data and FPL news current, caching results in Redis and PostgreSQL.
- Deployed on Railway via Docker with JWT authentication, multi-session chat history, and async PostgreSQL connection pooling.

Book Recommendation Assistant — Semantic Search Chatbot

Flask · Scrapy · Cohere · Pinecone · MySQL

- Scraped Amazon book data using Scrapy, storing structured records in MySQL and exporting to JSON for downstream processing.
- Generated vector embeddings from book metadata using Cohere's Embed API and indexed them in Pinecone for efficient semantic retrieval.
- Built a Flask-based chatbot interface (Bookie) that embeds user queries at runtime and performs semantic search against Pinecone to surface relevant book recommendations via Cohere's chat API.
- Designed a modular project structure separating scraping, embedding, database, and web app layers for maintainability and extensibility.

Education

2019 – 2024
Abeokuta, Nigeria

Bachelor of Engineering (B.Eng.) in Electrical and Electronics Engineering (First Class)

Federal University of Agriculture, Abeokuta.